

20-Class Plant Disease Object Detection

YOLOv8 + PrismGhost Backbone & Quad-Head Multi-Scale Detect
GitHub: <https://github.com/Pranava19/PRISM-Plant-Disease-Detection>

1. Executive Summary

This research develops a lightweight, edge-deployable deep learning framework for **20-Class Plant Disease Object Detection**. We enhanced the standard YOLOv8 architecture by introducing a novel **PrismGhost convolutional module** to minimize computational FLOPs while preserving fine spatial lesion patterns, paired with a **Quad-Head Detection network (P2, P3, P4, P5)** specifically designed to detect micro-scale leaf lesions that standard 3-head detectors fail to localize.

- **PrismGhost Module:** Intrinsic feature extraction paired with group-wise depthwise cheap convolutions, reducing parameters to **2.25M** and complexity to **10.8 GFLOPs**.
- **Quad-Head Detection:** Adds a high-resolution P2/4 scale (160x160 feature maps) for early-stage micro lesion spots.
- **Unified 20-Class Dataset:** Harmonized **28,268** annotated images from PlantVillage and PlantDoc into an 80/10/10 split.
- **Edge Deployment:** Exported dynamic ONNX model (**8.64 MB**) operating at **11.8 ms** (~85 FPS) on GPU.

2. Benchmark Quantitative Performance

Metric	Score	Technical Significance
Precision (P)	92.32%	High fidelity; exceptionally low false alarm rate on healthy leaves
Recall (R)	76.90%	Comprehensive detection coverage across varying leaf angles & occlusions
mAP @ 50	84.65%	High mean average precision at standard 0.5 IoU threshold
mAP @ 50-95	78.76%	Strict bounding-box localization tightness across IoU thresholds (0.50:0.95)
Model Parameters	2.25 Million	Significantly lighter than baselines (YOLOv8n: 3.2M, YOLOv8s: 11.2M)
Complexity (GFLOPs)	10.8 GFLOPs	Low computational cost suitable for low-power edge devices
Inference Latency	11.8 ms	Real-time edge throughput (~85 FPS) on NVIDIA GPU
Model Size (ONNX)	8.64 MB	Framework-independent ONNX graph ready for mobile/drone IoT

3. Unified 20-Class Target Taxonomy

ID	Target Class Name	Host Crop	Pathogen / Condition
0	Apple Scab	Apple (Malus domestica)	Fungal (Venturia inaequalis)
1	Apple Cedar Rust	Apple (Malus domestica)	Fungal (Gymnosporangium juniperi)
2	Apple Healthy	Apple (Malus domestica)	Healthy / Asymptomatic
3	Corn Gray Leaf Spot	Corn (Zea mays)	Fungal (Cercospora zeae-maydis)
4	Corn Common Rust	Corn (Zea mays)	Fungal (Puccinia sorghi)
5	Corn Northern Leaf Blight	Corn (Zea mays)	Fungal (Exserohilum turcicum)
6	Grape Black Rot	Grape (Vitis vinifera)	Fungal (Guignardia bidwellii)
7	Grape Healthy	Grape (Vitis vinifera)	Healthy / Asymptomatic
8	Pepper Bacterial Spot	Bell Pepper (Capsicum annuum)	Bacterial (Xanthomonas campestris)
9	Pepper Healthy	Bell Pepper (Capsicum annuum)	Healthy / Asymptomatic
10	Potato Early Blight	Potato (Solanum tuberosum)	Fungal (Alternaria solani)
11	Potato Late Blight	Potato (Solanum tuberosum)	Oomycete (Phytophthora infestans)

12	Tomato Bacterial Spot	Tomato (Solanum lycopersicum)	Bacterial (Xanthomonas spp.)
13	Tomato Early Blight	Tomato (Solanum lycopersicum)	Fungal (Alternaria solani)
14	Tomato Late Blight	Tomato (Solanum lycopersicum)	Oomycete (Phytophthora infestans)
15	Tomato Leaf Mold	Tomato (Solanum lycopersicum)	Fungal (Passalora fulva)
16	Tomato Septoria Leaf Spot	Tomato (Solanum lycopersicum)	Fungal (Septoria lycopersici)
17	Tomato Yellow Leaf Curl Virus	Tomato (Solanum lycopersicum)	Viral (Begomovirus / TYLCV)
18	Tomato Mosaic Virus	Tomato (Solanum lycopersicum)	Viral (Tobamovirus / ToMV)
19	Tomato Healthy	Tomato (Solanum lycopersicum)	Healthy / Asymptomatic

4. Key Architectural Insights & Defense Points

- Why PrismGhost?** Standard convolution creates high feature map redundancy. PrismGhost separates computation into primary intrinsic convolutions and cheap depthwise operations, reducing FLOPs by ~50% without loss of detection accuracy.
- Why Quad-Head Detection?** Early-stage foliar diseases are tiny spots (< 16x16 px). Standard 3-head detectors downsample images by 8x, 16x, and 32x, causing small feature loss. The P2/4 head (160x160) provides direct high-resolution feature localization.
- Edge Deployment:** The exported ONNX graph (8.64 MB) runs at 85 FPS on GPU and is directly compatible with TensorRT, OpenVINO, and ONNXRuntime on edge IoT devices.